POKHARA UNIVERSITY

Level: Bachelor Semester: Spring Year : 2025
Programme: BCA Full Marks : 100
Course: Database Management System (New) Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain the three levels of data abstraction in a DBMS with suitable examples. How does data independence relate to these levels? 7

   b) Explain the two-tier and three-tier database architecture with diagrams. 8

2. a) Draw an ER diagram for a hospital system with entities like Patient, Doctor, Appointment, and Room. Show appropriate relationships and cardinalities. 7

   b) Consider the following relational database. 8

   Student (SID, Name, Age, Dept)

   Course (CID, CName, Credits)

   Enroll (SID, CID, Grade)

   Write relational algebra expressions for the following queries:

   i. Find names of students who are enrolled in the course "Database Systems".

   ii. List all students (SID, Name) who are not enrolled in any course.

   iii. Retrieve courses (CID, CName) with more than 3 credits.

   iv. Find the total number of students in each department.

3. a) What is view? Explain the importance of views with its syntax. 7

   OR

   What is a JOIN? Explain the types of joins with syntax and examples.

   b) Consider the following relational schema: 8

   Employees (EmpID, Name, DeptID, Salary, JoinDate)

   Departments (DeptID, DeptName, Location)

Projects (ProjectID, ProjectName, Budget, DeptID)

Write SQL queries for:

i. Create the table Employees with appropriate constraints.
ii. Find the Name, Salary of employee of 'IT' department.
iii. Increase the salary of employees in the "IT" department by 10%.
iv. Delete all projects with a budget less than 40,000.

4. a) What are integrity constraints? Explain in detail. 7

b) Define domain, entity integrity, and referential integrity constraints with examples. 8

OR

What is denormalization? When and why is it used in database design?

5. a) What is Conflict Serializability? Explain any one lock-based protocol. 7

b) Define query optimization. Explain the need and steps in query optimization. 8

6. a) What is checkpoint and transaction rollback? Explain the algorithm of log-based recovery. 7

b) What are the advantages of ORM databases? Compare NoSQL databases with traditional relational databases. 8

7. Write short notes on: (**Any two**) 2×5

a) Failure Classifications

b) Database instances and schemas

c) Generalization VS Aggregation